

Real-Time AQI Trends and Pollution Analysis



Project Overview and Objective

This project involves cleaning, transforming, and analysing raw Pollutant data using **Excel** and creating an interactive **Power BI dashboard** to derive meaningful insights.

Rapid urbanization and industrial development in India have resulted in rising air pollution levels. Although pollutant data is collected from multiple monitoring stations, it is not easy to interpret trends, compare regions, or identify major pollutant contributors. Without proper visualization and analysis, policymakers and the public cannot take timely action.

The main objective of this project is to analyse real-time air pollution data collected from various monitoring stations across Indian cities, identify major pollutant contributors, calculate and assess air quality patterns over time, and visualize pollution hotspots through interactive dashboards and geographic mapping in Power BI.

Data Sources

- **Source Description and Timeline:** <https://www.data.gov.in/resource/real-time-air-quality-index-various-locations> and 2025
- **Domain:** Environment

Problem Statement

- To calculate the Air Quality Index (AQI) using pollutant concentration values.
- To identify the major contributing pollutant at each monitoring station (PM2.5, PM10, NO₂, SO₂, CO, NH₃, OZONE).
- To compare pollutant levels across different states and cities in India.
- To create interactive dashboards that show regional pollution trends, AQI categories, and hotspot locations using latitude & longitude.
- To provide insights that support environmental awareness and help authorities take preventive measures.

Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Country	String (Text)	Country name
State	String (Text)	State name
City	String (Text)	City name
Station	String (Text)	Monitoring station name
Date and Time	Date/Time	Timestamp of recorded data
Latitude	Decimal number	Geographic latitude coordinate
Longitude	Decimal number	Geographic longitude coordinate
Pollutant_id	String (Text)	Pollutant type(PM2.5, PM10, NO ₂ , SO ₂ , CO, NH ₃ , OZONE)
Pollutant_min	Integer	Minimum recorded pollutant value
Pollutant_max	Integer	Maximum recorded pollutant value
Pollutant_avg	Integer	Average pollutant value
AQI value	Integer	Air Quality Index value
AQI category	String (Text)	Categories(Good, Moderate,Satisfactory, poor, very poor, severe)

Tools & Technologies

- **Excel:** Data cleaning and transformation.
 - **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.
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Data Pre-Processing

- **Data Collection:**
Gathered data from Data.gov.in (18-11-2025)
- **Data Cleaning & Transformation:** (Power Query)
Removed duplicates, handled missing values and standardized formats.
- **Data Transformation:**
Created conditional column (AQI category) based on the column created using DAX (AQI value)

- **Filtering & Sorting:**

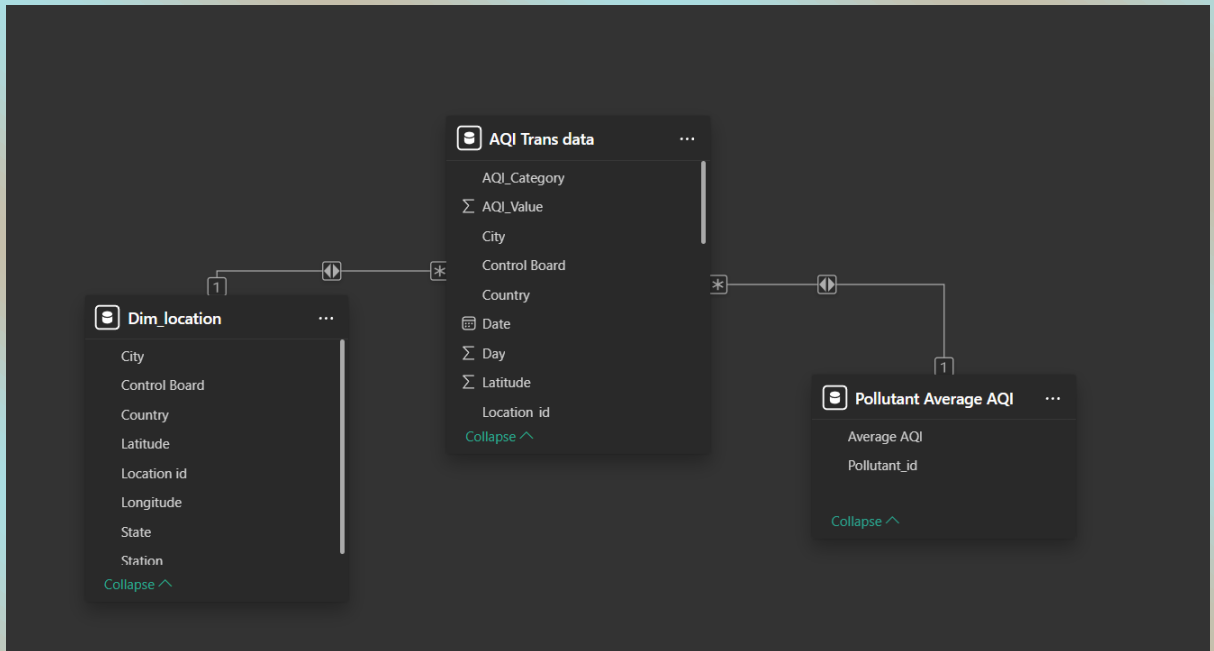
Organized data to focus on relevant records.

- **Data Integration:**

Unified the cleaned data set for further visualization

Data Modelling and DAX (Power BI)

- **Data Model:** Established relationships between tables and defined cardinality.



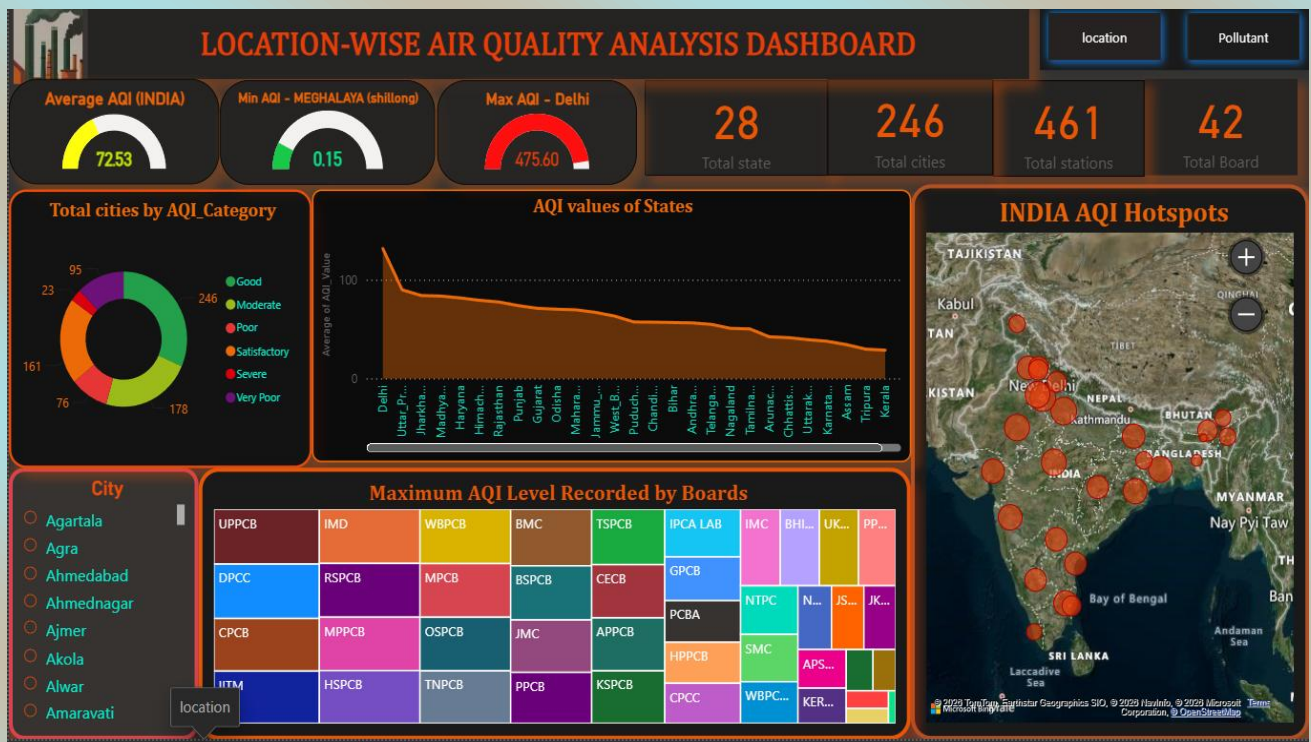
- **Calculated Columns & DAX Measures:** Implemented DAX formulas for Calculated column – **AQI Value** and for key metrics such as Average AQI, Min AQI, Max AQI, Total states, Total cities, Total stations and Total boards.

Analysis and Visualizations (Power BI)

Developed interactive Power BI dashboards addressing all problem statements using:

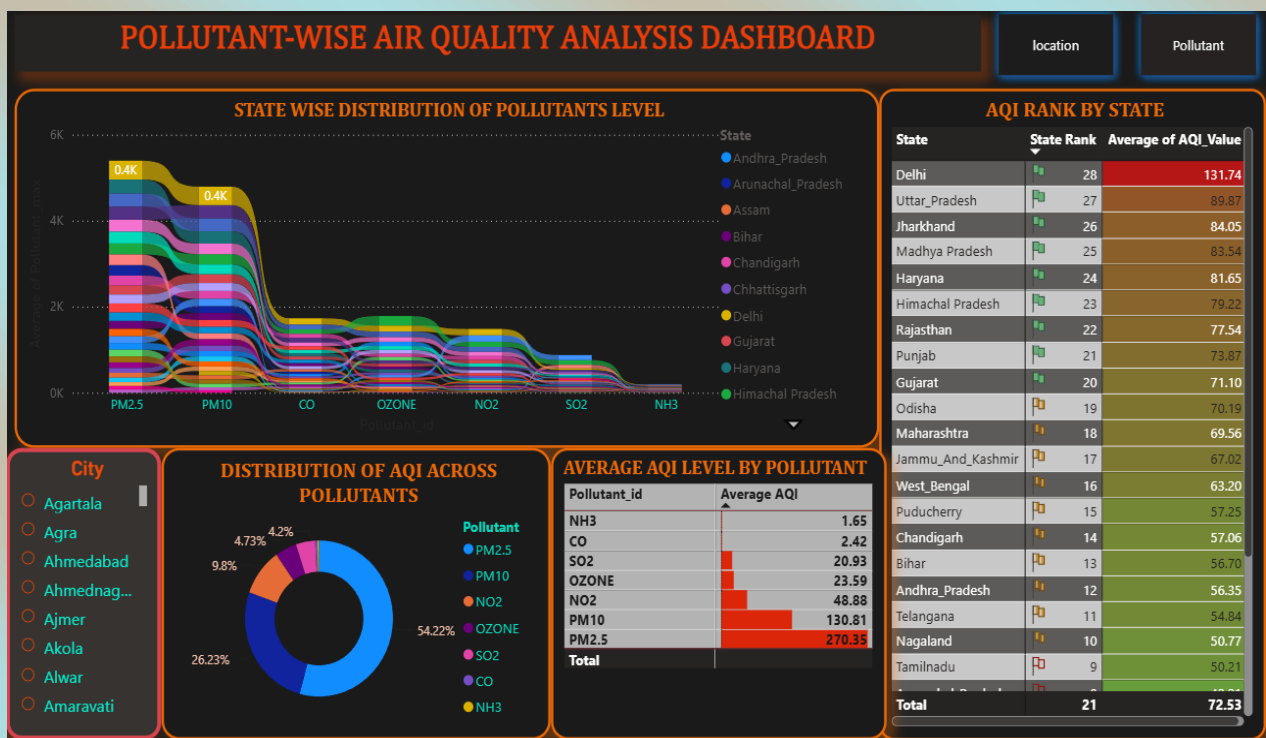
Visual 1:

- **Donut Chart:** AQI distribution across different category.
- **Stacked area chart:** AQI trend by state
- **Tree Maps:** Control boards records based on AQI recorded in hierarchy
- **Map visual:** Displays **INDIA AQI** hotspot
- **Cards:** Displays Average, minimum and maximum AQI , Helps to quickly identify cleanest and most polluted states.



Visual 2:

- **Donut chart:** It visualizes how each pollutant contributes to national AQI share.
- **Matrix table:** States are ranked based on pollution level along with AQI values.
- **Ribbon chart:** Shows pollutant intensity across states and determine which state struggle with specific pollutants.
- **Table visual:** shows which pollutant contribute most to air pollution.



Insights :

Descriptive :

- Overall air quality varies significantly across states, with some cities consistently showing high pollutant concentrations, especially PM2.5 and PM10.
- PM2.5 and PM10 are the most dominant pollutants recorded across the monitoring stations, contributing the maximum share to AQI.
- Several cities maintain good AQI levels, indicating effective environmental regulations and lower industrial activities.

Diagnosis:

- High PM2.5 and PM10 levels are mainly due to vehicular emissions, construction dust, and industrial output, causing poor AQI in dense urban areas.
- Increased NO₂ levels correlate with peak traffic periods, revealing transportation as the main contributor.
- Regions with low AQI variation have better pollution control mechanisms, such as green cover and fewer emission sources.

Predictive:

- If current trends continue, PM2.5 and PM10 will remain the primary contributors to air pollution in most major cities.
- AQI values are expected to worsen during winter months, when calm winds and low temperatures trap pollutants.
- Cities currently showing moderate AQI may slip into poor category if emissions continue at the same rate without intervention.

Prescriptive:

- Reduce PM2.5/PM10 emissions by enforcing dust control at construction sites and regulating vehicular emissions.
 - Promote public transportation and restrict peak-hour traffic, especially in cities with recurrent AQI spikes.
 - Install more air-quality monitoring stations in high-risk zones to improve real-time pollution control decisions.
 - Increase green cover and promote tree plantation drives, which help absorb pollutants and improve air quality.
 - Issue public advisories during high-AQI periods, especially for children, elders, and asthma patients.
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Conclusions

The analysis of air quality dataset across multiple Indian states and monitoring stations reveals significant variation in pollution levels based on geography, pollutant type, and time.

- By examining pollutant concentration values such as PM2.5, PM10, NO₂, SO₂, CO, NH₃, and OZONE, it is evident that particulate matter (especially PM2.5 and PM10) and nitrogen dioxide remain the most common contributors to poor air quality in many regions.
- Geographic mapping using latitude and longitude further visualizes clusters of polluted areas, allowing for better environmental planning and resource allocation.
- Overall, this analysis demonstrates the importance of continuous air-quality monitoring, the need for region-specific pollution control measures, and the value of data-driven dashboards in improving environmental awareness. T
- The insights generated can assist policymakers, environmental agencies, and the public in understanding pollution patterns and taking informed actions to improve air quality.